

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Case Study

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA15

COURSE OUTCOME COVERED

CO3: *Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications. (BL3)*

Assessment Tool Weightage: 20%

Dave's Taxonomy: Precision (Level 3)
Question Count: 3

Total Marks: 30

Code of Conduct

I JAGADISH.V(192521352) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: JAGADISH.V

Assessment Tasks

Case Study

1. Case 1: A food delivery company wants to deploy a cloud-native backend with web APIs, browser clients, and shared cloud storage for receipts and delivery images. Propose the architecture and explain the role of each component.
2. Case 2: A media startup must decide between SAN-backed storage, object storage, and serverless processing for video upload workflows. Analyze the case and recommend a deployment model.
3. Case 3: A smart city platform needs low-latency decision making near sensors while still using the cloud for analytics. Analyze the use of edge computing and fog computing in this environment.

Case Study Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Case	25%	Clearly identifies all technical and business requirements	Identifies most requirements	Basic understanding only	Poor understanding of case
Application of Concepts	25%	Applies suitable cloud concepts and tools effectively	Good application with minor issues	Partial application	Weak or incorrect application
Analysis	20%	Strong analysis with clear trade-off reasoning	Reasonable analysis	Limited analysis	Minimal analysis
Solution Design	20%	Solution is feasible, practical, and well justified	Reasonable solution with minor gaps	Basic solution	Unclear solution
Clarity	10%	Well-structured and easy to follow	Mostly clear	Somewhat unclear	Poorly organized

Case 1: A food delivery company wants to deploy a cloud-native backend with web APIs, browser clients, and shared cloud storage for receipts and delivery images. Propose the architecture and explain the role of each component

A: Role of Each Component

1. **Web Browser / Mobile Client**

Customers use the browser or mobile application to browse restaurants, order food, make payments, and track deliveries.

2. **API Gateway / Web API**

Receives requests from clients and routes them to the appropriate backend service. It provides a common entry point for REST APIs.

3. **Cloud-Native Backend**

Handles core business operations such as restaurants, menus, orders, payments, delivery tracking, and order status.

4. **Authentication Service**

Verifies users through login and controls access to customer, restaurant, and delivery accounts.

5. **Cloud Database**

Stores structured information such as customer details, restaurant information, menus, orders, payments, and delivery status.

6. **Cloud Object Storage**

Stores large files such as **food images, delivery images, invoices, and receipts**. Object storage is suitable because these files can be stored and accessed independently.

7. **Monitoring Service**

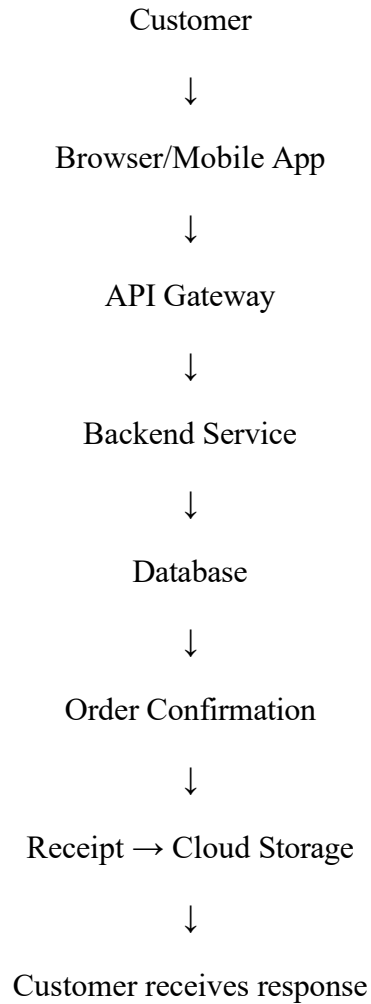
Monitors API performance, errors, server health, and application availability. Alerts administrators when problems occur.

Example Cloud Services

The architecture could use:

- **AWS API Gateway** → API management
- **AWS ECS/Lambda** → Backend
- **Amazon RDS/DynamoDB** → Database
- **Amazon S3** → Receipts and delivery images
- **Amazon CloudWatch** → Monitoring

WORKING FLOW



Conclusion

This cloud-native architecture provides **scalability, security, availability, and centralized storage**. The web/API layer handles user requests, the backend manages food delivery operations, the database stores application data, and cloud object storage securely stores receipts and delivery images.

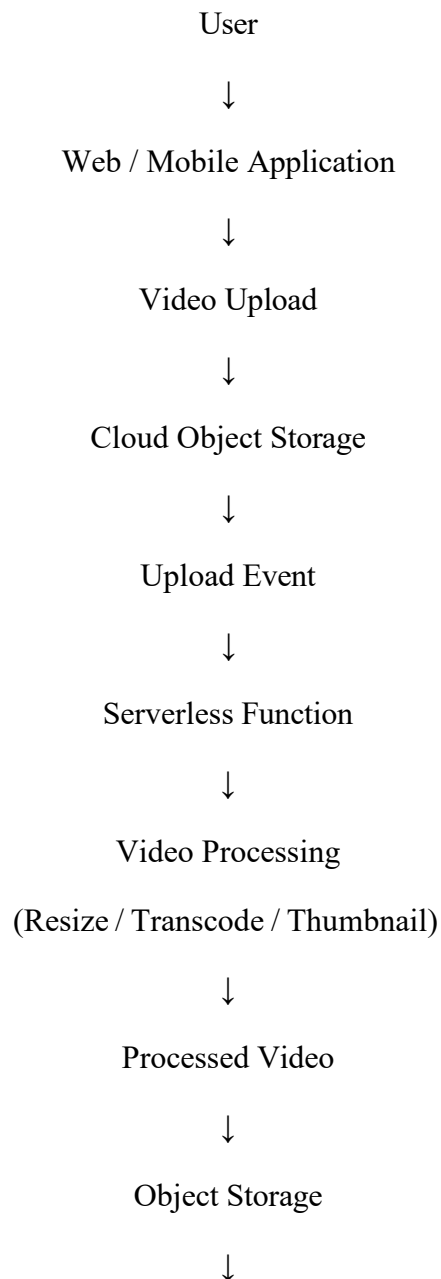
2. A media startup must decide between SAN-backed storage, object storage, and serverless processing for video upload workflows. Analyze the case and recommend a deployment model.

A: Problem:

A media startup needs to store and process a large number of video files. It must choose between **SAN-backed storage, object storage, and serverless processing**.

Option	Advantages	Disadvantages	Suitability
SAN-backed Storage	Very high performance, low latency	Expensive and requires infrastructure management	Medium
Object Storage	Highly scalable, durable, cost-effective	Slightly higher access latency than SAN	Excellent
Serverless Processing	Automatic scaling, no server management, pay-per-use	Not ideal for storing videos itself	4. Excellent for processing

Architecture



Working:

1. The user uploads a video through the media application.
2. The original video is stored in **cloud object storage**.
3. The upload automatically generates an event.
4. A **serverless function** is triggered.
5. The function performs tasks such as video transcoding, compression, or thumbnail generation.
6. The processed files are stored back in object storage.
7. Users can access or stream the processed videos.

Why Not SAN?

SAN-backed storage provides excellent performance but is more expensive and requires more infrastructure management. For a startup handling potentially **large and unpredictable video volumes**, object storage provides better scalability and cost efficiency.

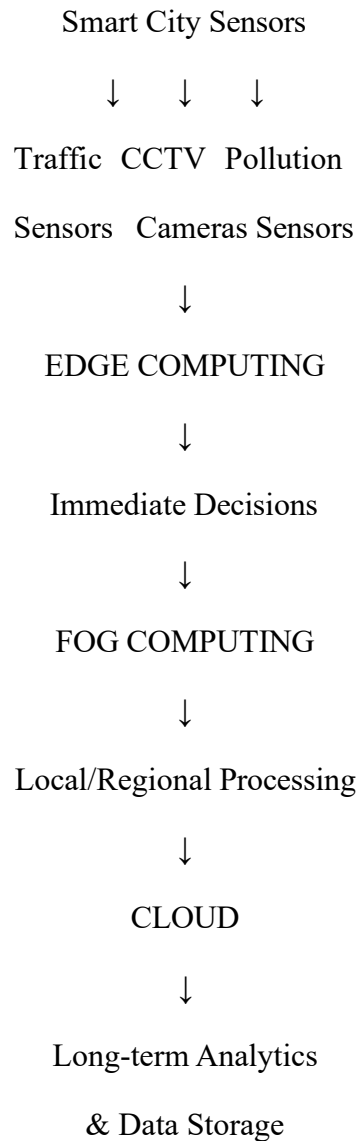
Example Cloud Services

- **Object Storage:** Amazon S3 / Azure Blob Storage
- **Serverless:** AWS Lambda / Azure Functions
- **Video Processing:** AWS Elemental MediaConvert / similar cloud media service
- **Monitoring:** Amazon CloudWatch / Azure Monitor

3: Smart City – Edge and Fog Computing

A: A smart city has many sensors collecting real-time data from **traffic lights, CCTV cameras, parking systems, pollution sensors, and public transport**. The system needs quick decisions near the sensors while using the cloud for large-scale analytics.

Proposed Architecture



Edge Computing:

Edge computing processes data directly near the sensors or devices. For example, a traffic camera can detect heavy traffic and immediately send a command to change the traffic signal.

Benefits:

- Very low latency
- Faster decision making
- Reduces data sent to the cloud
- Works even with limited connectivity

Example:

A road sensor detects an accident → edge device immediately alerts the nearby traffic signal system.

A local fog gateway can collect information from many traffic sensors and cameras in a particular area.

Benefits:

- Processes data locally before sending it to the cloud
- Reduces network bandwidth
- Coordinates multiple nearby devices
- Provides low-latency regional decisions

Example:

A fog gateway receives traffic information from 20 nearby intersections and coordinates their traffic signals.

Role of Cloud Computing

The cloud is used for tasks that do not require immediate responses:

- Long-term data storage
- City-wide traffic analysis
- Machine learning
- Pollution trend analysis
- Historical reports
- Predictive analytics